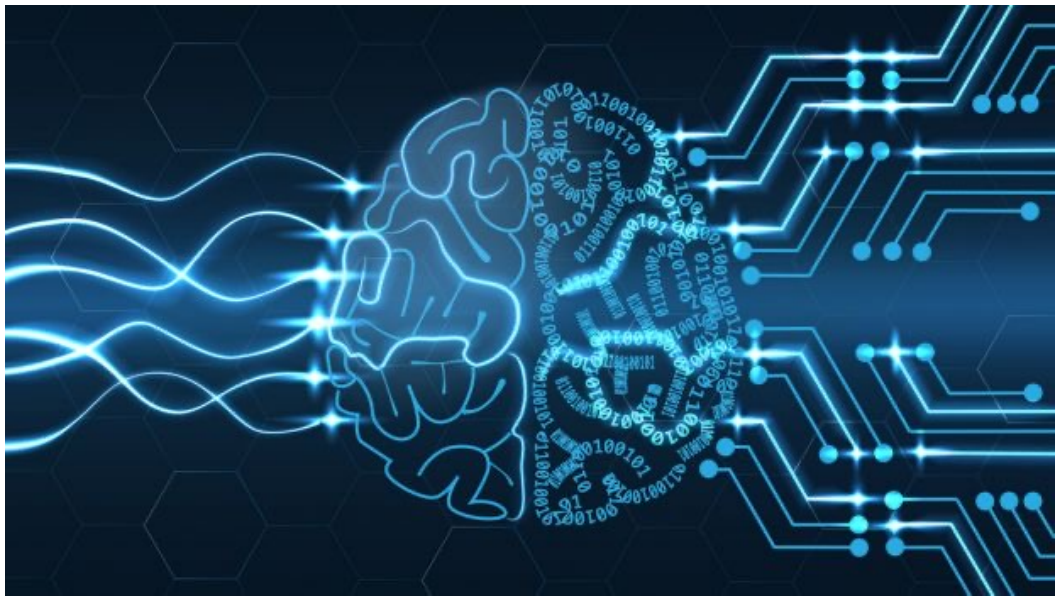

Chapter 3. Machine Learning with Probability Models

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Department of Systems and Computer Engineering

Outline

- Statistical Learning Basics
- Maximum Likelihood Estimator
- Logistic Regression

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- Statistical Learning Basics
- Maximum Likelihood Estimator
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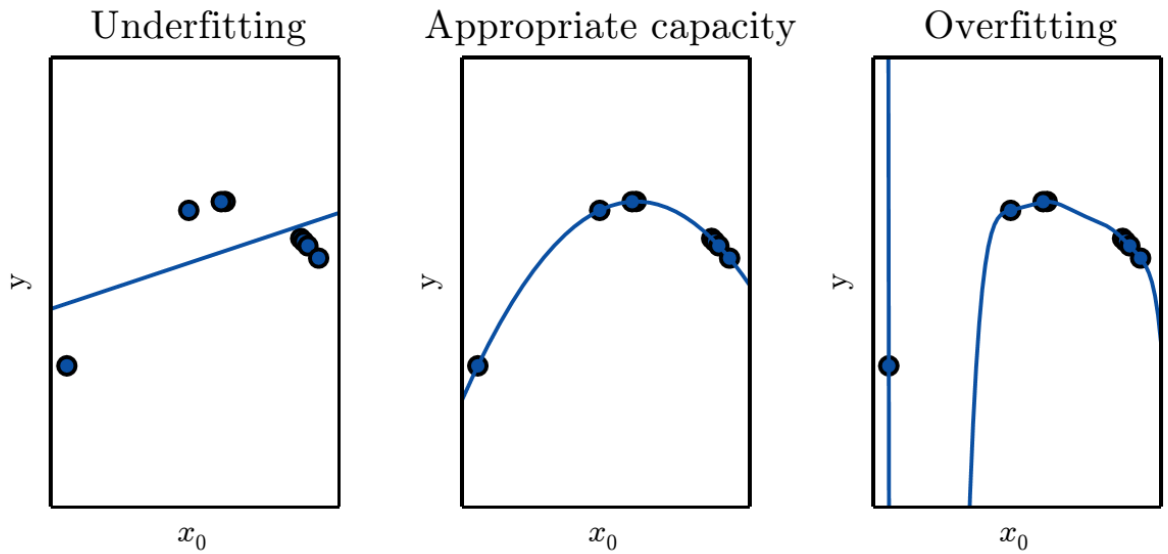
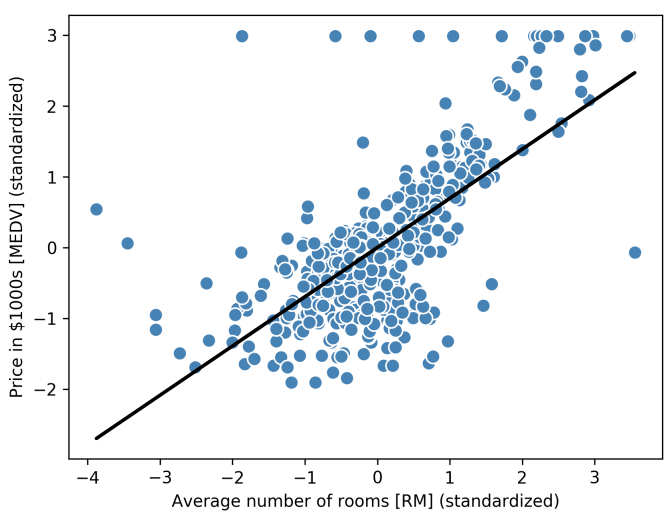
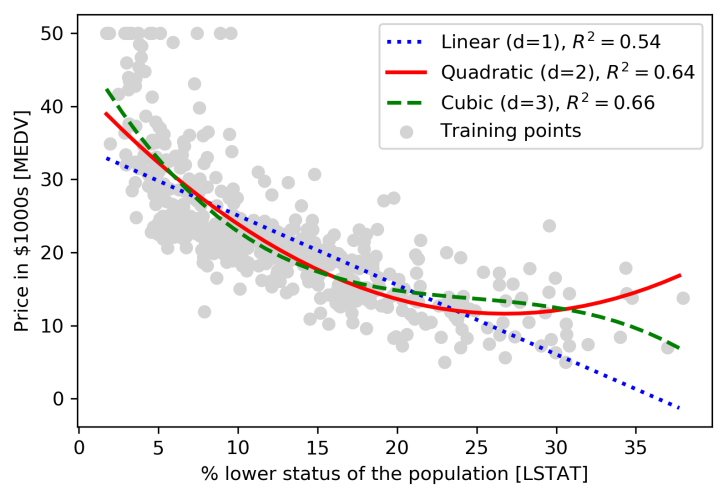
Why statistical learning?







Why statistical learning?

- Why random?
  - Inherent randomness in the system being modeled
  - Incomplete observability
  - Incomplete modeling
- Why statistical learning?
  - Better capacity
  - Better representations
  - Better performance estimation
    - Benchmark
    - Bounds

Probability Review

- Strong Law of Large Numbers
  - Let $X_1, X_2, \cdots, X_n$ be a sequence of IID with mean μ AND variance σ^2, $\hat{X}_n = \frac{X_1+X_2+\cdots+X_n}{n}$ .Then, with probability 1, $\hat{X}_n \to \mu$, as $n \to \infty$ and $\mathrm{VAR}(\hat{X}_n) = \mathbb{E}\left[\left(\hat{X}_n - \mu\right)^2\right] = \frac{\sigma^2}{n}$

    Note: Standard error decreases with $\sqrt{n}$
  - In general, $Y = g(X)$, $Y_i = g(X_i)$, $\hat{Y}_n = \frac{g(X_1)+g(X_2)+\cdots+g(X_n)}{n}$

    then $\lim_{n\to\infty} \hat{Y}_n = \mathbb{E}[Y] = \int_{-\infty}^{\infty} g(x)p(x)dx$, $\mathrm{VAR}(\hat{Y}_n) = \frac{1}{n}VAR(g(X))$
- Monte Carlo Principle

$$I_A(X) = \begin{cases} 1, if\ X \in A \\ 0, if\ X \notin A \end{cases} \qquad \hat{Y}_n(A) = \frac{\sum_{i=1}^{n} I_A(X_i)}{n} \to \mathbb{E}[I_A(X)] = p(A)$$

  - A way to estimate probability

$$p(a) = \int_A f(x)dx$$

  - A way to estimate an integral

Probability Review

- Conditional probability and chain rule

$$p(Y|X) = \frac{p(X,Y)}{p(X)}, \quad p(X,Y) = p(X)p(Y|X)$$

$$p(X_1, X_2, \cdots, X_N) = p(X_1) \prod_2^N p(X_i|X_1, \cdots, X_{i-1})$$

- Independence

$$p(Y|X) = p(Y) \quad p(X,Y) = p(X)p(Y)$$

- Marginal distribution and total probability rule

$$p(Y) = \sum_x p(x,Y) = \sum_x p(Y|x)p(x)$$

- Bayes' formula

a priori

$$p(X|Y) = \frac{p(X)p(Y|X)}{p(Y)} = \frac{p(X)p(Y|X)}{\sum_x p(Y|x)p(x)}$$

a posteriori

Statistical Learning Basics

- Supervised learning
  - Given sample points $(x_i, y_i), i = 1, \cdots, M$
  - Find distribution $p(Y|X)$ (compare with $f(X)$ in ch2)
    - For learning, assume $p(Y|X:\theta)$
  - Find optimal $\hat{\theta}$ with $(x_i, y_i), i = 1, \cdots, M$
  - Prediction with $\hat{y} = \underset{y}{\operatorname{argmax}}\, p\left(Y \middle| X:\hat{\theta}\right)$
- Unsupervised learning
  - Given sample points $x_i, i = 1, \cdots, M$
  - Find distribution $p(X, H)$
  - Assume $p(X, H:\theta)$
  - Find $p(H|X:\theta')$ and $p(X|H:\theta'')$
  - Find optimal $\hat{\theta}', \hat{\theta}'', \hat{\theta}$ with $x_i, i = 1, \cdots, M$

Understanding Errors

- Assume for $X \in \mathbb{R}^N, Y \in \mathbb{R}$ is random variables with $\mathbb{E}(Y|X) = \mu(x), \mathrm{Var}(Y|X) = \sigma^2(x)$, $Y_1|X, Y_2|X, \cdots, Y_M|X$ IID
- Look for a function $\hat{Y} = f(X, \hat{\theta})$, $\hat{\theta} = g(Y_1, Y_2, \cdots, Y_M)$
- Minimize MSE (Mean Squared Error)

$$MSE = \mathbb{E}\left(\left[Y - \hat{Y}\right]^2 | X\right) = \mathbb{E}([Y - f(x, \theta)]^2 | X)$$

$$f(x, \hat{\theta}) = \underset{\hat{Y}}{\mathrm{argmin}}\, \mathbb{E}(\left[Y - \hat{Y}\right]^2 | X)$$

$$= \underset{\hat{Y}}{\mathrm{argmin}}\, \mathbb{E}(\left[Y - \mathbb{E}(Y) + \mathbb{E}(Y) - \hat{Y}\right]^2)$$

$$= \mathbb{E}[Y - \mathbb{E}(Y)]^2 + \underset{\hat{Y}}{\mathrm{argmin}}(\left[\mathbb{E}(Y) - \hat{Y}\right]^2) + 2\mathbb{E}\left[(Y - \mathbb{E}(Y))\left(\mathbb{E}(Y) - \hat{Y}\right)\right]$$

$$= \sigma^2(x) + \underset{\hat{Y}}{\mathrm{argmin}}(\left[\mathbb{E}(Y) - \hat{Y}\right]^2)$$

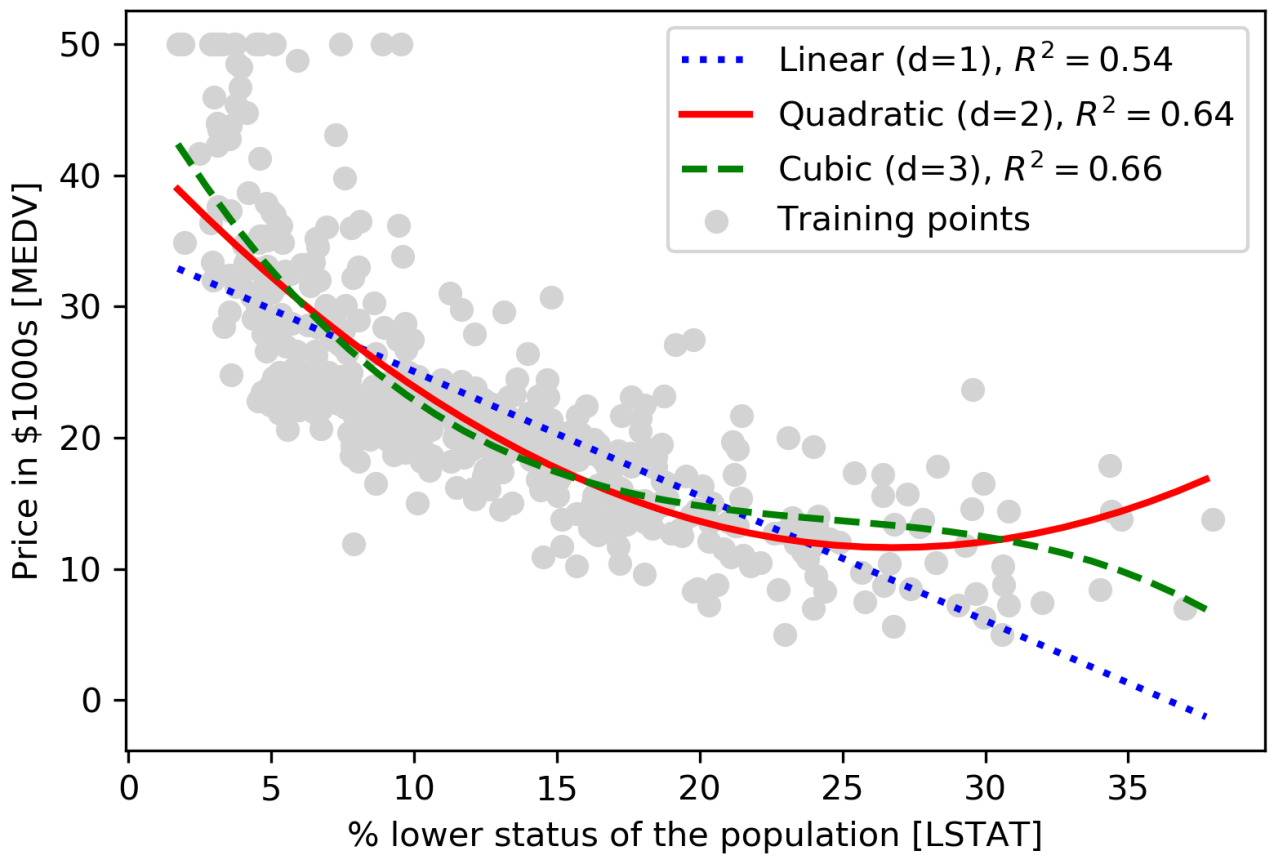
intrinsic error

model error

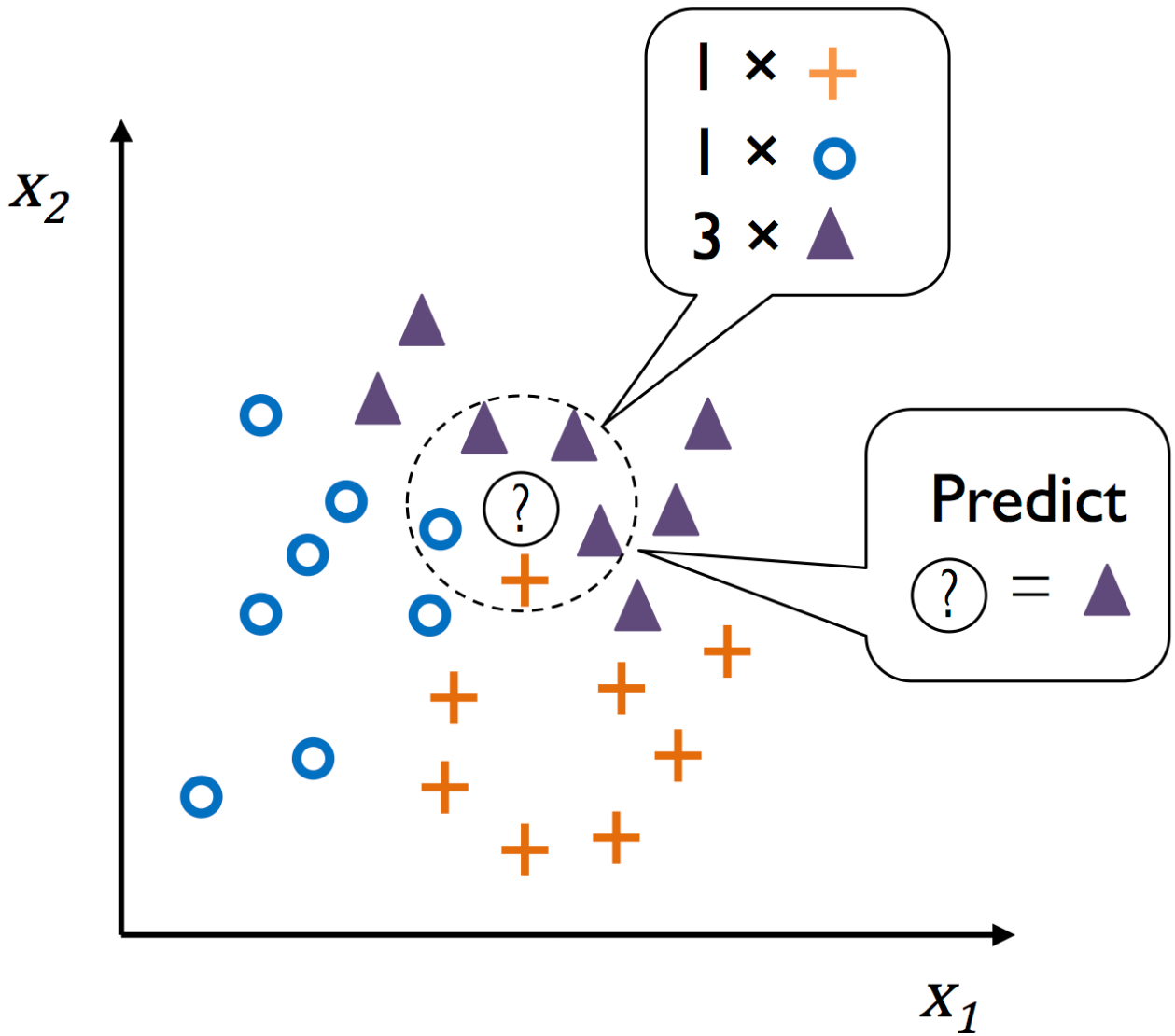
Optimal model:

$$\hat{Y} = f(x, \hat{\theta}) = \mathbb{E}(Y) = \mu(x)$$

An example


Linear (d=1), $R^2 = 0.54$
Quadratic (d=2), $R^2 = 0.64$
Cubic (d=3), $R^2 = 0.66$
Training points
Price in $1000s [MEDV]
% lower status of the population [LSTAT]
0
10
20
30
40
50
0
5
10
15
20
25
30
35

K-Nearest Neighbors



KNN

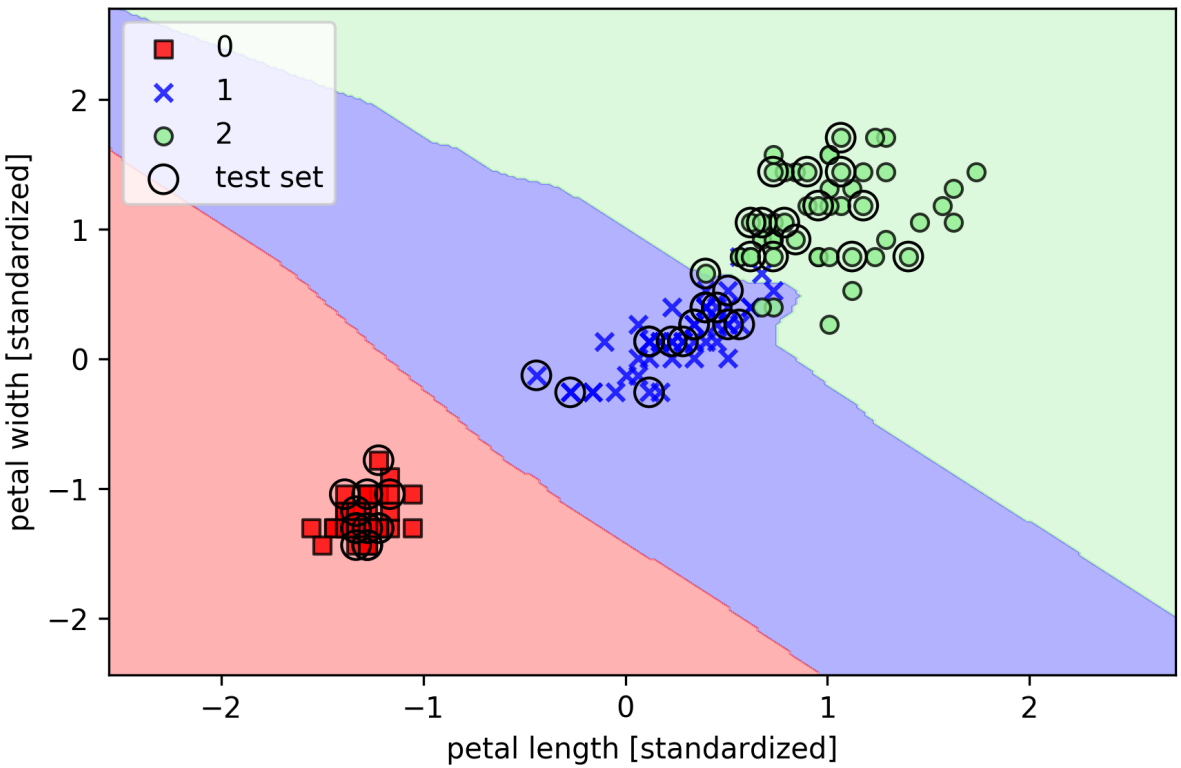
- K-nearest neighbor
  - A nonparametric model
  - Assumption: local constancy
  - Classification
    - Given a query point x, we find the $x_i, i = 1, \cdots, K$ closest in distance to x, then classify using majority vote among the K neighbors

$$\hat{Y} = f(x) = \operatorname*{argmax}_c \frac{\sum_{x_i} I_{x_i \in N_k(x), c}}{K}$$

  - Regression

$$\hat{Y} = f(x) = \frac{1}{k} \sum_{x_i \in N_k(x)} y_i$$

Example


0
1
2
test set
petal width [standardized]
petal length [standardized]
−2
−1
0
1
2

The Curse of Dimensionality

- The number of examples required goes up exponentially
  - $O(v^N)$ for N dimensions with v values in each dimension

Prior Assumptions

- Implicit assumptions
  - Local constancy and smoothness
- Explicit assumptions
  - Functions and parameters

An Example

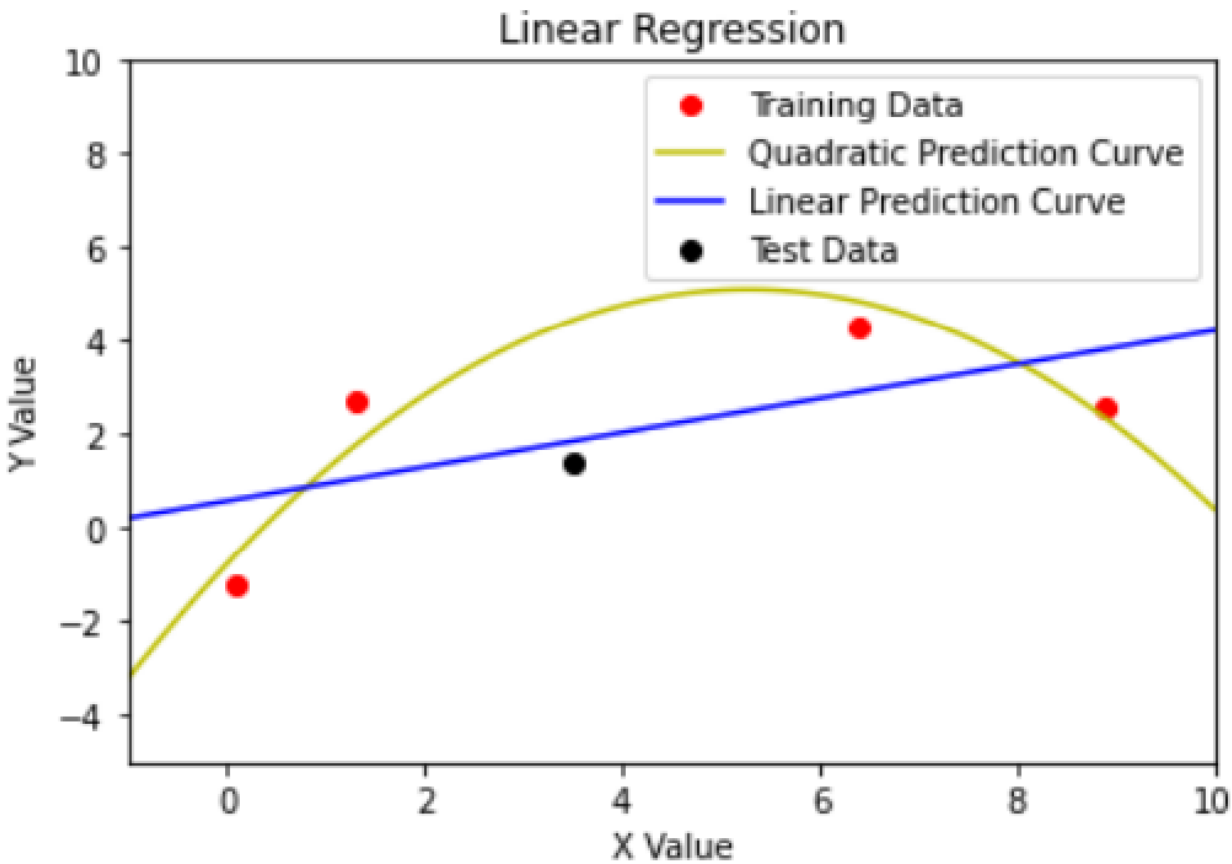
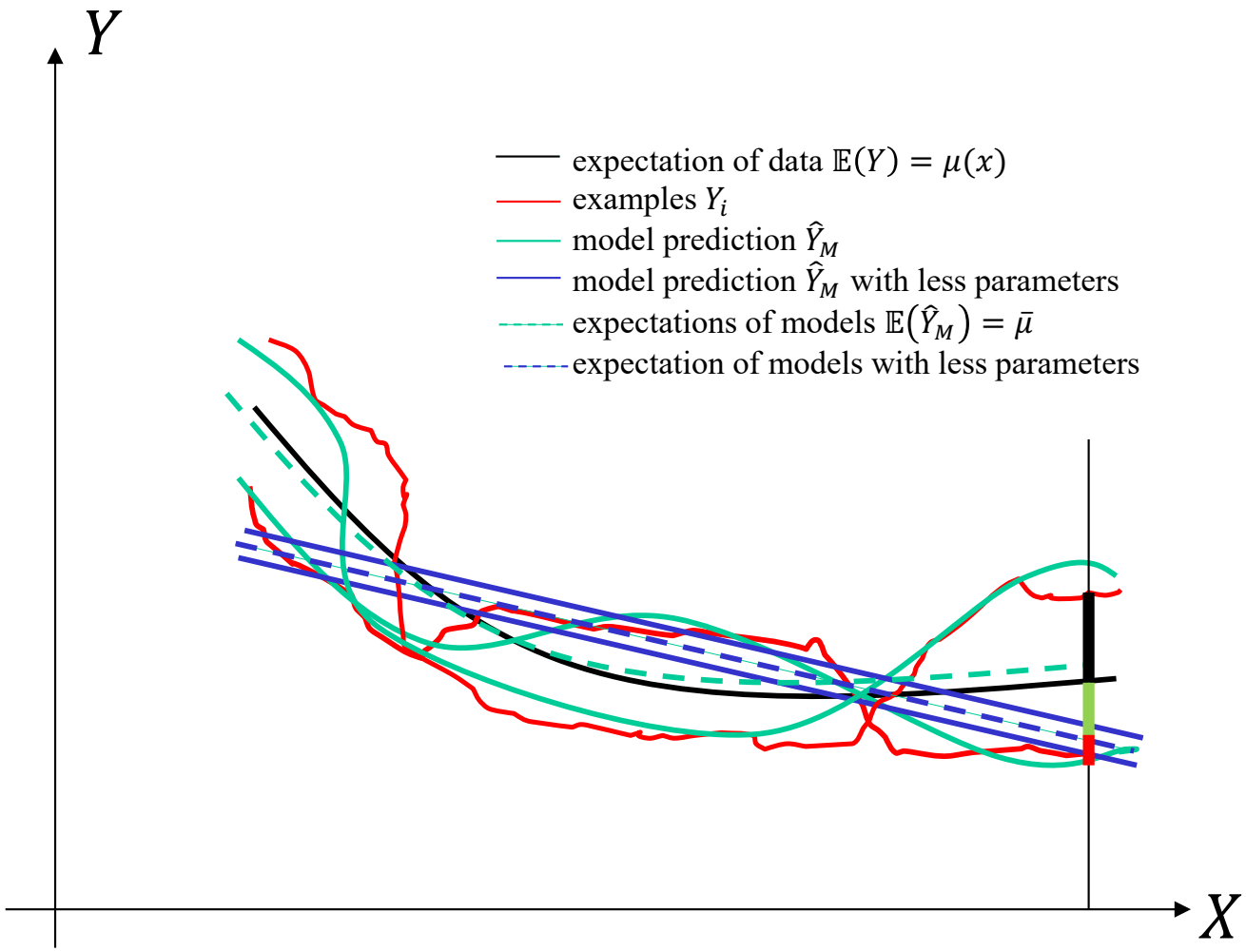

Linear Regression
Training Data
Quadratic Prediction Curve
Linear Prediction Curve
Test Data
Y Value
X Value
10
8
6
4
2
0
−2
−4
0
2
4
6
8
10

Illustration of Errors



Understanding Errors

- Model Errors

$Y: \mu(x), \sigma^2(x)$

$\hat{Y}_M = f(X, Y_1, \dots, Y_M), \mathbb{E}(\hat{Y}_M|X) = \bar{\mu}(x),\ \mathrm{Var}(\hat{Y}_M|X) = \bar{\sigma}^2(x)$

$B(\hat{Y}_M|X) = \mathbb{E}(\hat{Y}_M|X) - \mu(x) = \bar{\mu}(\mathrm{x}) - \mu(x)$

$$\mathbb{E}\left[\left(\mathbb{E}(Y) - \hat{Y}_M\right)^2|X\right] = \mathbb{E}\left[\left(\mu(x) - \hat{Y}_M\right)^2\right]$$

$$= \mathbb{E}\left[\left(\mu(x) - \mathbb{E}\hat{Y}_M + \mathbb{E}\hat{Y}_M - \hat{Y}_M\right)^2\right]$$

$$= \mathbb{E}\left[\left(\mu(x) - \mathbb{E}\hat{Y}_M\right)^2\right] + \mathbb{E}\left[\left(\mathbb{E}\hat{Y}_M - \hat{Y}_M\right)^2\right] + 2\mathbb{E}[\left(\mu(x) - \mathbb{E}\hat{Y}_M\right)\left(\mathbb{E}\hat{Y}_M - \hat{Y}_M\right)]$$

$$= \mathbb{E}\left[\left(\mu(x) - \mathbb{E}\hat{Y}_M\right)^2\right] + \mathbb{E}\left[\left(\mathbb{E}\hat{Y}_M - \hat{Y}_M\right)^2\right]$$

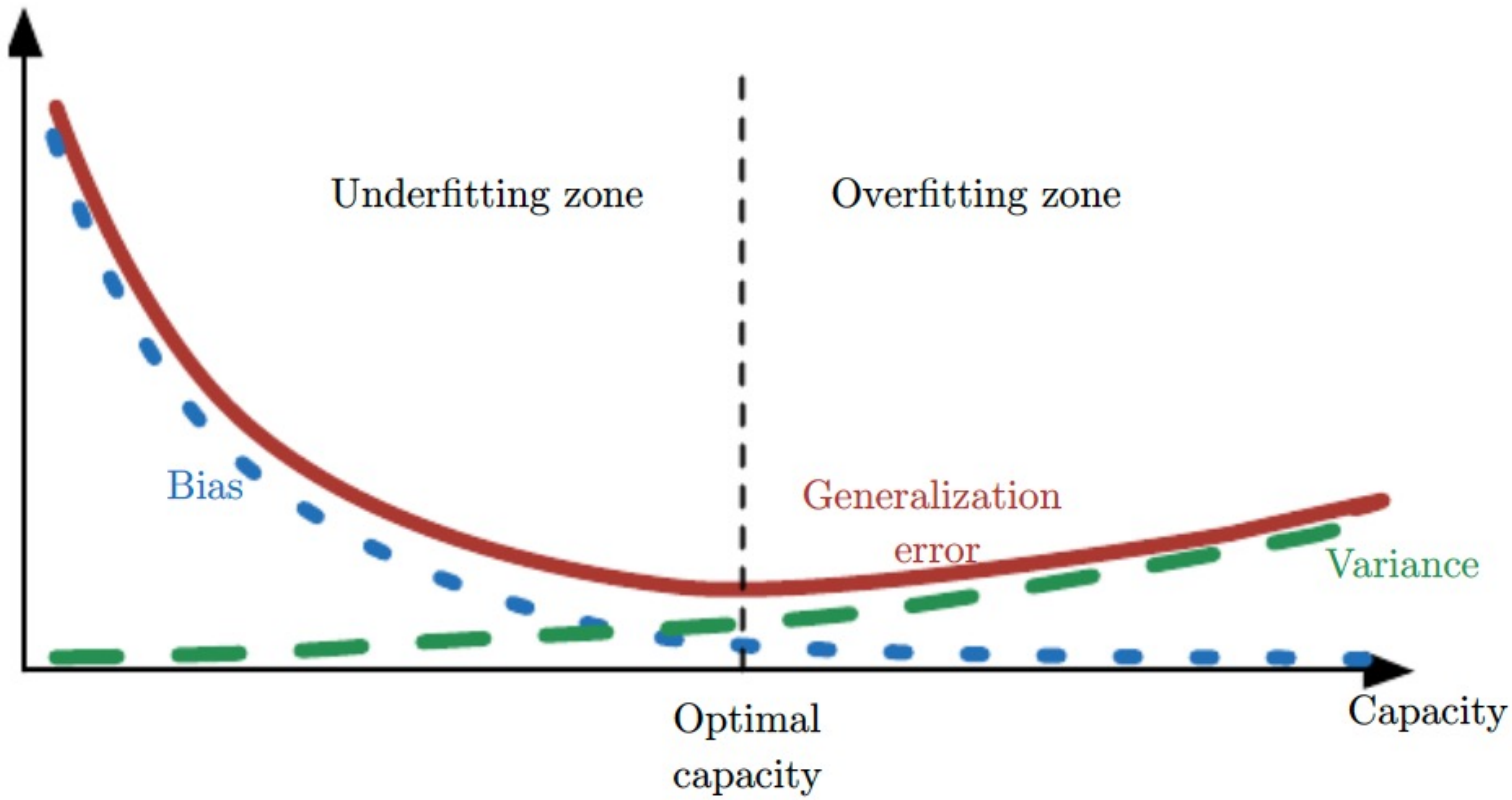
$$= (\mu(x) - \bar{\mu}(x))^2 + \bar{\sigma}^2(x)$$

$$= \left[B\left(\hat{Y}_M\right)\right]^2 + \bar{\sigma}^2(x)$$

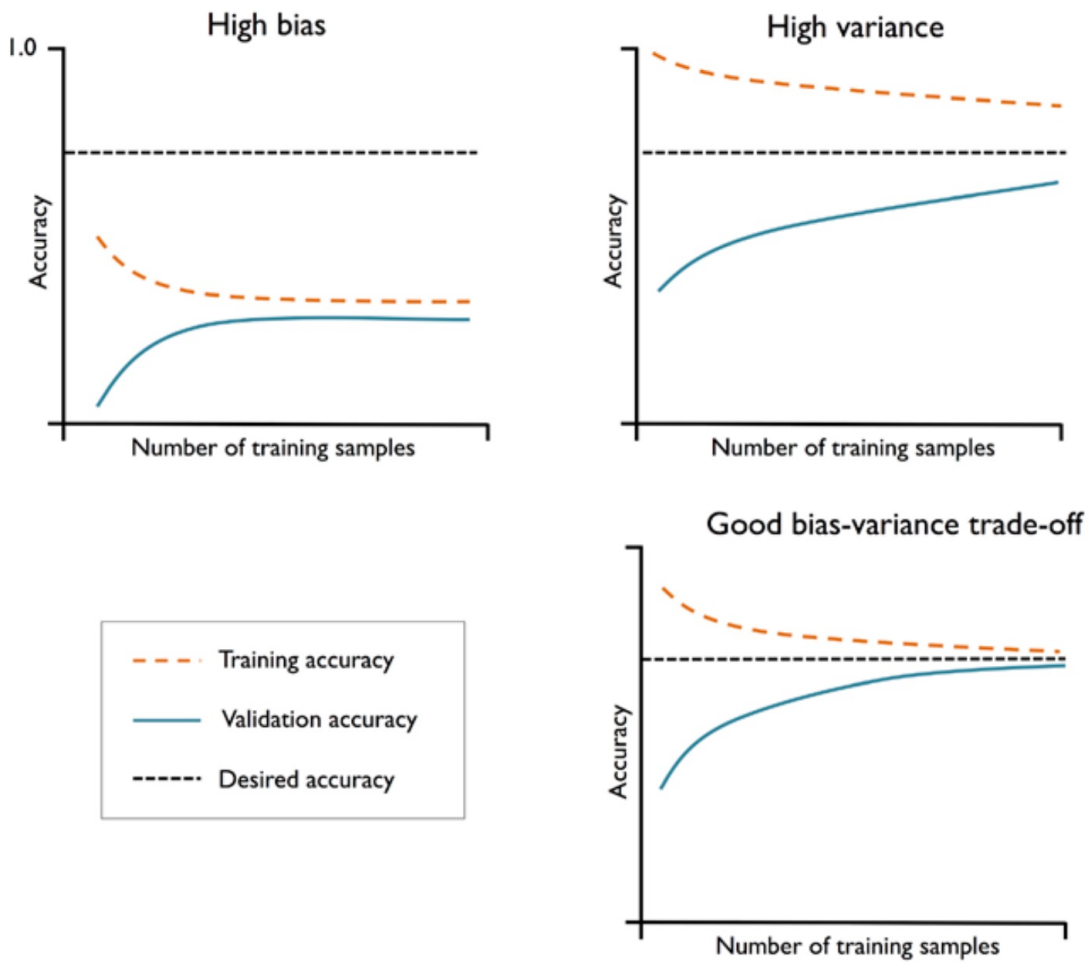
model bias

model variance

Errors vs. Capacity



Selecting the right capacity



Mitigating Errors

- Intrinsic Errors
  - Increase input dimensionality
- Model Bias
  - Increase model capacity
- Sample Variance
  - Increase data size

Outline

- Statistical Learning Basics
- Maximum Likelihood Estimator
- Logistic Regression

LSE Problem

- Prediction based on probability

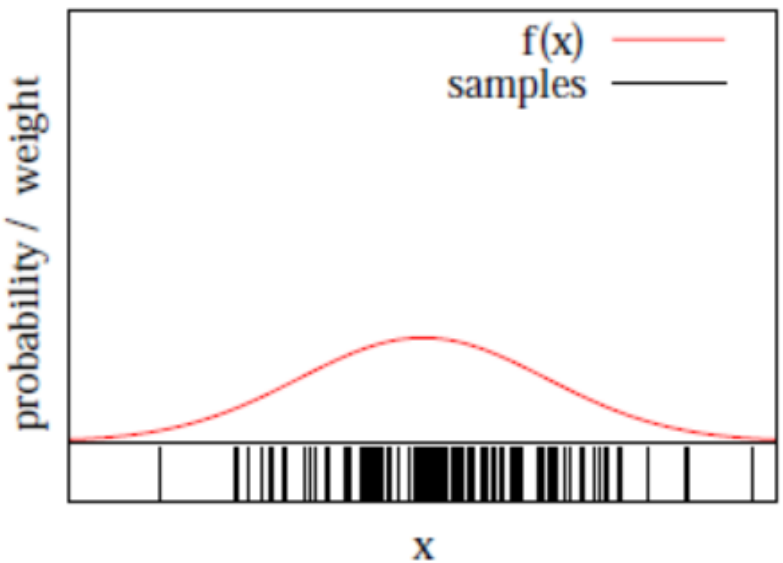
$$\hat{y} = \underset{y}{\operatorname{argmax}}\, p(Y|X:\theta)$$

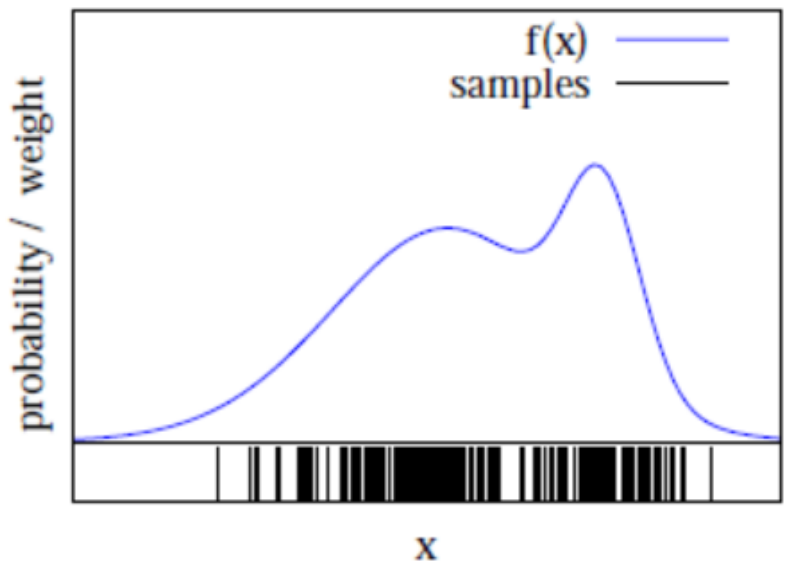
- Hard to work for categorical applications

$$L(\boldsymbol{\theta}) = \frac{1}{M}\left[\sum_{i=1}^{M}\left(y_i - \underset{y}{\operatorname{argmax}}\, p\left(y:\boldsymbol{x}^{(i)},\boldsymbol{\theta}\right)\right)^2\right]$$

  - Discrete RV leads to step function
  - Step function leads to useless gradient

LSE Problem





- An alternative

$$L(\boldsymbol{\theta}) = \frac{1}{M}\left[\sum_{i=1}^{M}\left(1 - p\left(y_i : \boldsymbol{x}^{(i)}, \boldsymbol{\theta}\right)\right)^2\right]$$

  - Hard to optimize: often exponential function

Training with Statistical Models

- Objective of training
  - Need a new objective to bypass prediction of Y
  - Assume data come from a model with θ
  - Goal: find $\hat{\theta}_M$ close to θ
  - Challenge: no labels for θ
- Properties of point estimator
  - Bias: $\mathrm{Bias}(\hat{\theta}_M) = \mathbb{E}(\hat{\theta}_M) - \theta$
  - Asymptotically unbiased: $\lim_{M\to\infty} \mathrm{Bias}(\hat{\theta}_M) = 0$
  - Mean squared error (MSE)

$$\mathrm{MSE}(\hat{\theta}_M) = \mathbb{E}\left[\left(\hat{\theta}_M - \theta\right)^2\right] = \mathrm{Bias}(\hat{\theta}_M)^2 + \mathrm{Var}(\hat{\theta}_M)$$

  - Consistency

$$\operatorname*{plim}_{M\to\infty} \hat{\theta}_M = \theta$$

Maximum Likelihood Estimator

- Maximum Likelihood Estimator (MLE)
  - A more general way for point estimator
  - Principle of MLE: the most reasonable values for θ are those for which the probability of the observed sample is largest

$$\hat{\theta}_{ml} = \underset{\theta}{\operatorname{argmax}}\, p_{model}(\boldsymbol{y}|\boldsymbol{x};\theta)$$

  - Assume y_i are independent sample points

$$\hat{\theta}_{ml} = \underset{\theta}{\operatorname{argmax}} \prod_{i=1}^{M} p_{model}(y^{(i)}|x^{(i)};\theta)$$

$$= \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^{M} \log p_{model}(y^{(i)}|x^{(i)};\theta)$$

  - Avoid gradient saturation for exponential functions

MLE Example

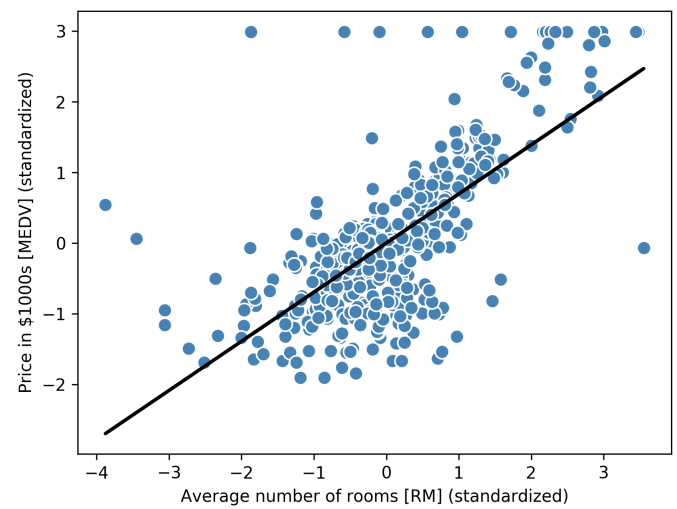
$$p(y|x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{y - m(\boldsymbol{x}:\boldsymbol{\beta})}{\sigma}\right)^2\right]$$

$$\log f(y_1, y_2, \cdots, y_M) = -\frac{1}{2}\sum_{i=1}^{M}\left(\frac{y_i - m(\boldsymbol{x}_i:\boldsymbol{\beta})}{\sigma}\right)^2 - \frac{1}{2}M\log\sigma^2 + C = l(\boldsymbol{\beta}, \sigma^2)$$

$$m(\boldsymbol{x}:\boldsymbol{\beta}) = \boldsymbol{x}^T\boldsymbol{\beta}$$

$$\nabla_\beta\, l(\boldsymbol{\beta}, \sigma^2) = 0 \qquad \widehat{\boldsymbol{\beta}} = (X^TX)^{-1}X^T\boldsymbol{y}$$

$$\frac{\partial l(\widehat{\boldsymbol{\beta}}, \sigma^2)}{\partial \sigma^2} = 0 \qquad \sum_{i=1}^{M}\frac{1}{2}\frac{\left(y_i - \boldsymbol{x}_i^T\widehat{\boldsymbol{\beta}}\right)^2}{\sigma^4} - \frac{M}{2\sigma^2} = 0$$



$$\hat{\sigma}^2 = \frac{1}{M}\sum_{i=1}^{M}\left(y_i - \boldsymbol{x}_i^T\widehat{\boldsymbol{\beta}}\right)^2 \qquad \hat{\sigma}^2 = \frac{1}{M}\sum_{i=1}^{M}\left(y_i - \boldsymbol{x}_i^T\widehat{\boldsymbol{\beta}}\right)^2$$

Properties of MLE

- Conditions
  - The true distribution must lie within the model family
  - The true distribution must correspond to exactly one value of θ
- Properties of MLE
  - Consistency

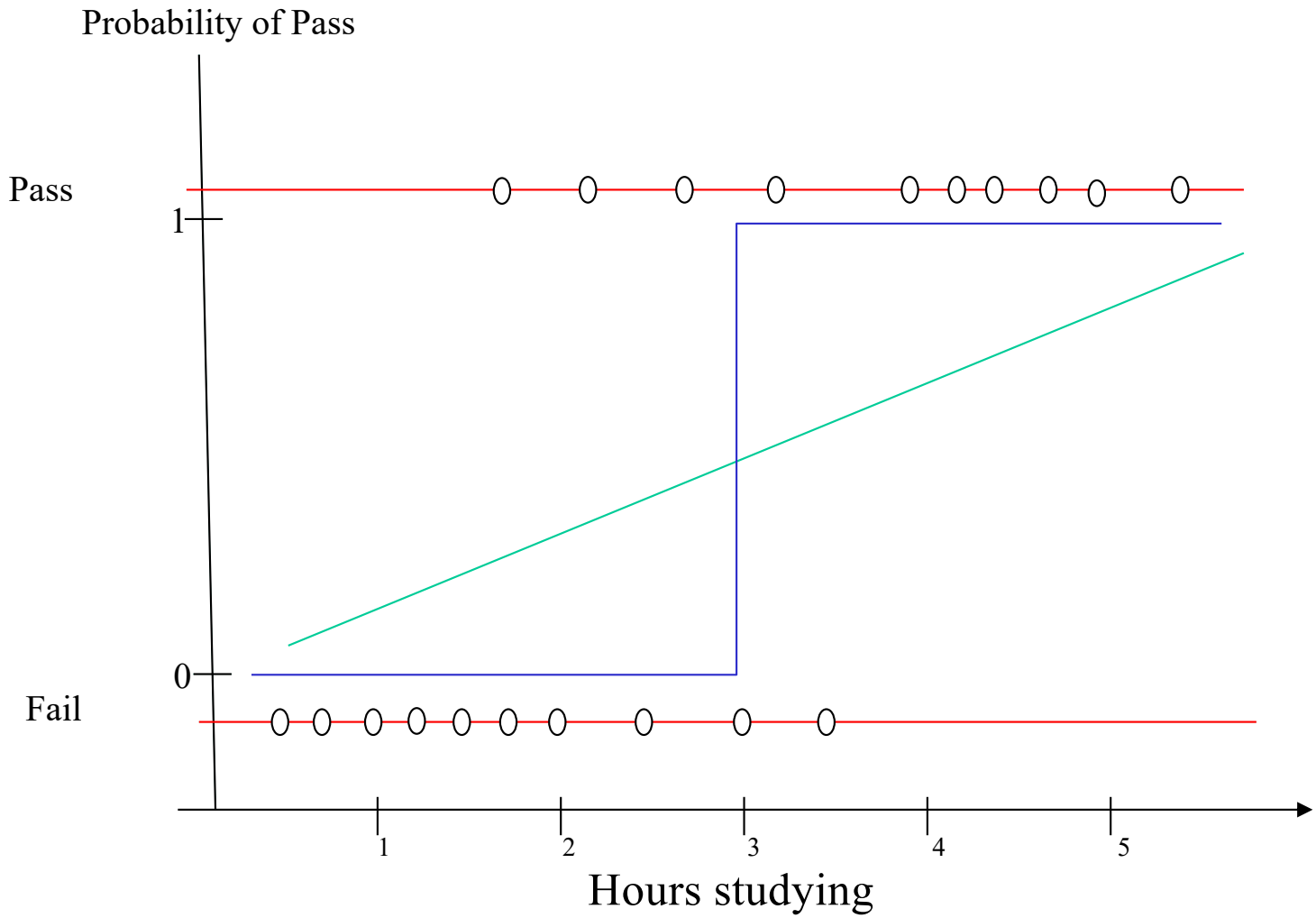
$$\operatorname*{plim}_{M \to \infty} \hat{\theta}_M = \theta$$

  - Efficiency: No consistent estimator has lower MSE of $\hat{\theta}_M$ than MLE

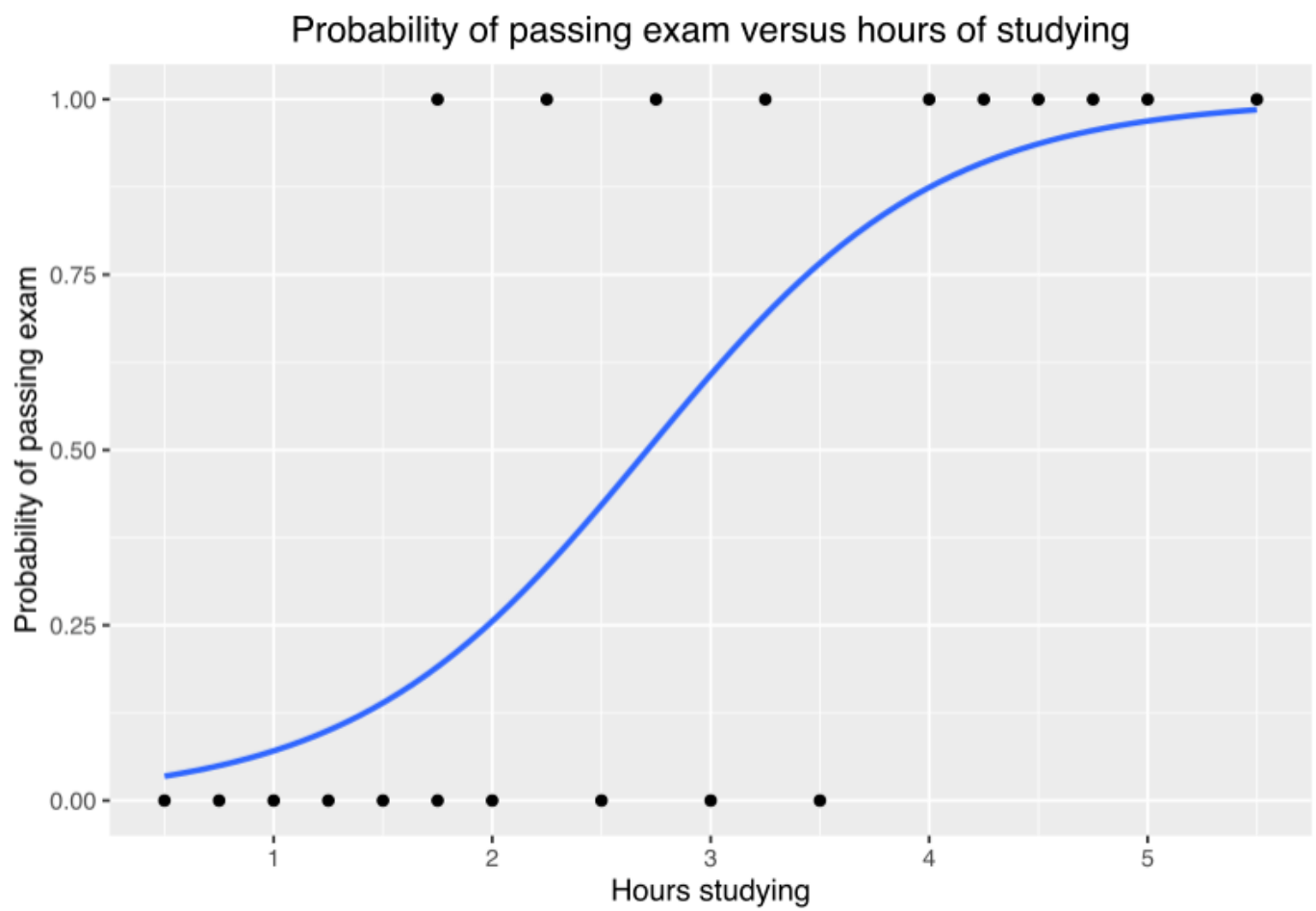
Outline

- Statistical Learning Basics
- Maximum Likelihood Estimator
- Logistic Regression

Example: Student Passing Exam

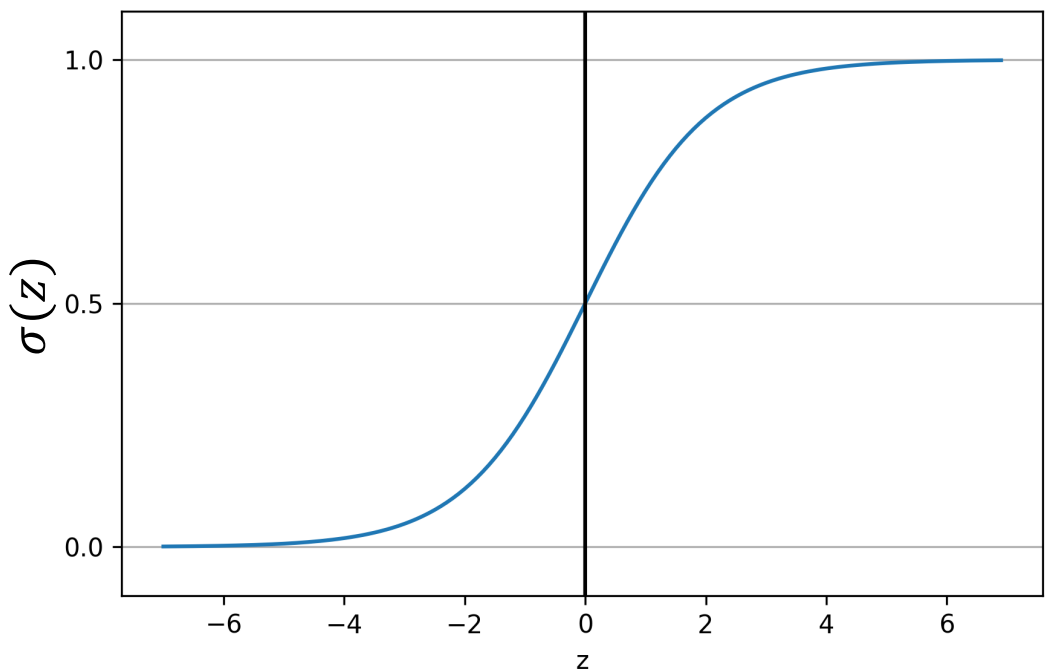


Objective


Probability of passing exam versus hours of studying
Probability of passing exam
1.00
0.75
0.50
0.25
0.00
1
2
3
4
5
Hours studying

Logistic Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}} = \frac{e^{z}}{1 + e^{z}}$$



Logistic Regression Model

- For $K = 2$ classes

$$p_1(x) = p(Y = 1|x), p_2(x) = p(Y = 2|x)$$
$$p_1(x) + p_2(x) = 1$$

  - Let

$$z = \beta_0 + \beta_1 x$$

- Assume a model family

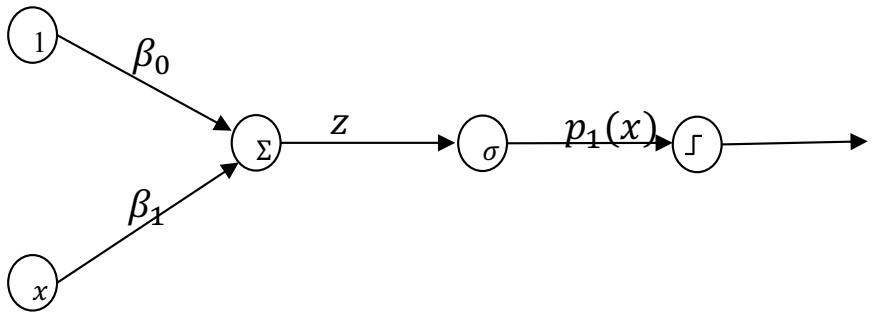
$$p_1(x) = \sigma(z)$$
$$= \frac{1}{1 + e^{-(\beta_0+\beta_1 x)}} = \frac{e^{(\beta_0+\beta_1 x)}}{1 + e^{(\beta_0+\beta_1 x)}}$$
$$p_2(x) = 1 - \sigma(z)$$
$$= \frac{e^{-(\beta_0+\beta_1 x)}}{1 + e^{-(\beta_0+\beta_1 x)}} = \frac{1}{1 + e^{(\beta_0+\beta_1 x)}}$$

- Property

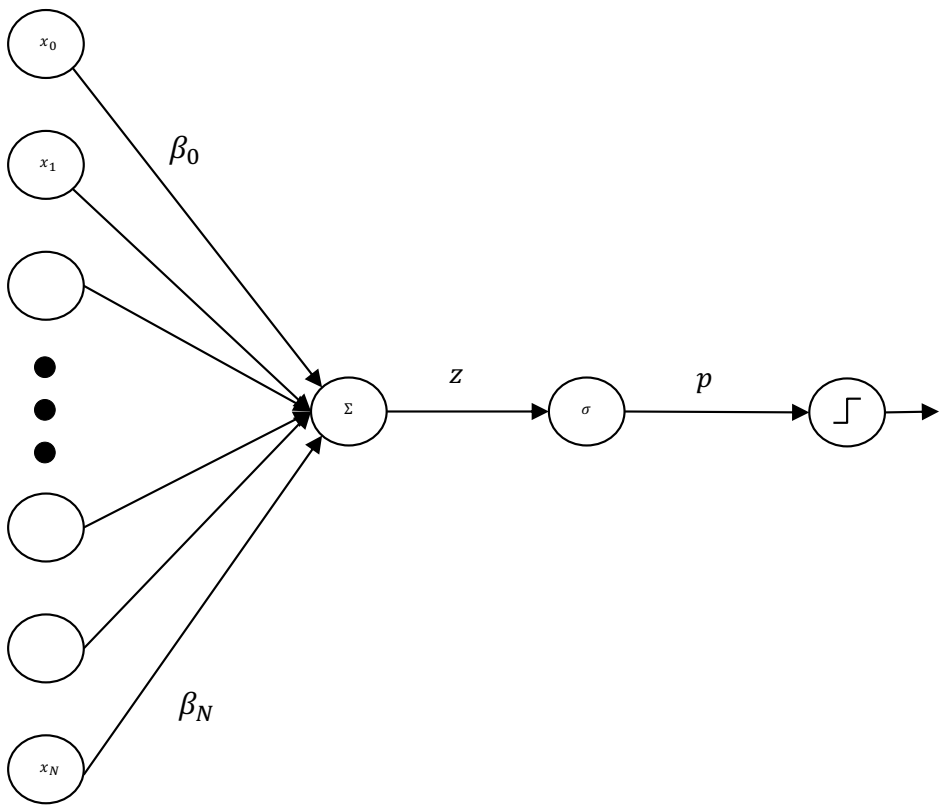
$$\text{logit}\big(p_1(x)\big) = \log \frac{p_1(x)}{1 - p_1(x)} = \log e^{(\beta_0+\beta_1 x)} = \beta_0 + \beta_1 x$$
$$\beta_0 + \beta_1 x > 0,\ p_1(x) > p_2(x)$$
$$\beta_0 + \beta_1 x = 0,\ p_1(x) = p_2(x)$$
$$\beta_0 + \beta_1 x < 0,\ p_1(x) < p_2(x)$$

Prediction

One dimension input



N dimension input



Training for Logistic Regression Model

- Log-likelihood for M observations

$$L(\beta) = \sum_{i=1}^{M} \log p\left(y^{(i)} \middle| x^{(i)}, \beta\right)$$

- For $K = 2$,

Probability selection

$$L(\beta) = \sum_{i=1}^{M} \left[I_1(y^{(i)}) \log p_1(x^{(i)}) + I_2(y^{(i)}) \log p_2(x^{(i)})\right]$$

where $I_k(.)$ is an indicator function

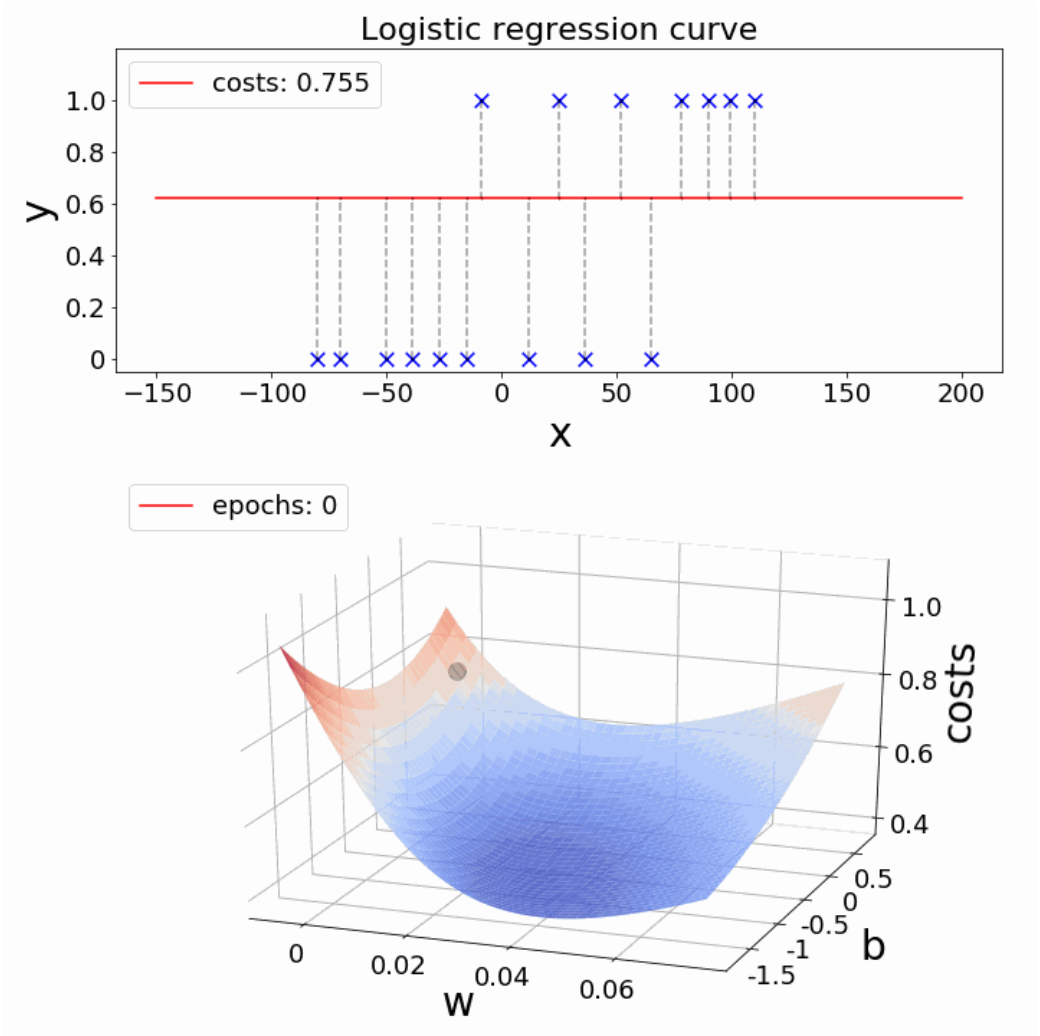
$$= \sum_{i=1}^{M} \left[I_1(y^{(i)}) \log\left(\sigma(z^{(i)})\right) + I_2(y^{(i)}) \log(1 - \sigma(z^{(i)}))\right]$$

$$J(\beta) = -\sum_{i=1}^{M} \left[I_1(y^{(i)}) \log\left(\sigma(z^{(i)})\right) + I_2(y^{(i)}) \log(1 - \sigma\left(z^{(i)}\right))\right]$$

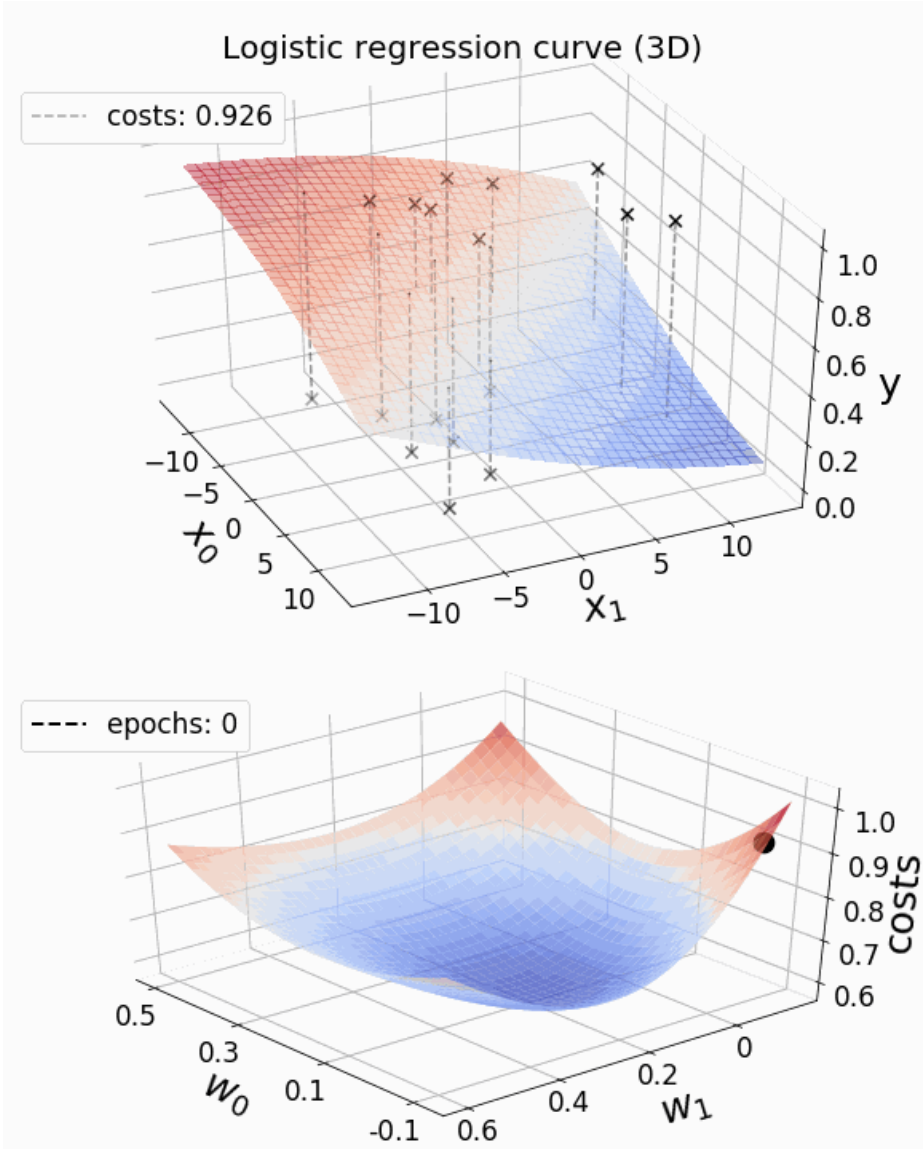
A convex function

$$\nabla_\beta J(\beta) = -\sum_{i=1}^{M} \left[I_1(y^{(i)}) \left(1 - \sigma\left(z^{(i)}\right)\right) x^{(i)} - I_2(y^{(i)}) \sigma\left(z^{(i)}\right) x^{(i)}\right]$$

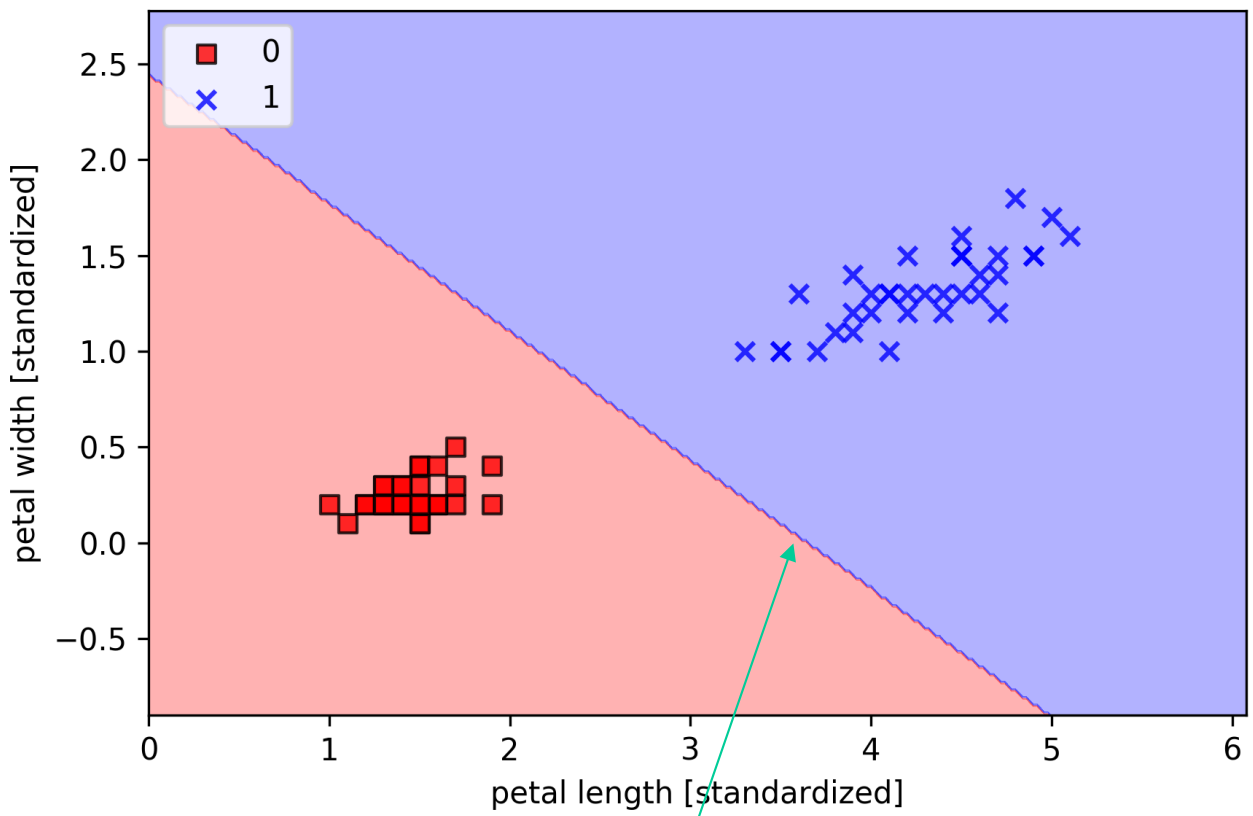
Example: One-Dimensional Input



Example: Two-Dimensional Input



Example



$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 = 0$$

An Alternative Notation

- Denote class 1 as $Y = 1$ and class 2 as $Y = 0$

$$L(\beta) = \sum_{i=1}^{M} \left[y^{(i)} \log p_1(x^{(i)}) + (1 - y^{(i)}) \log p_2(x^{(i)}) \right]$$

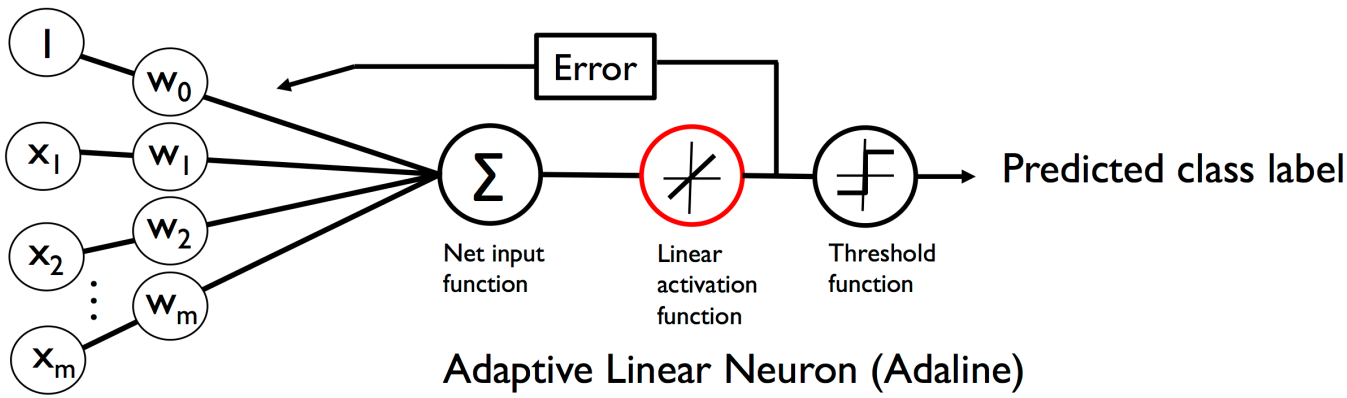
- Loss function

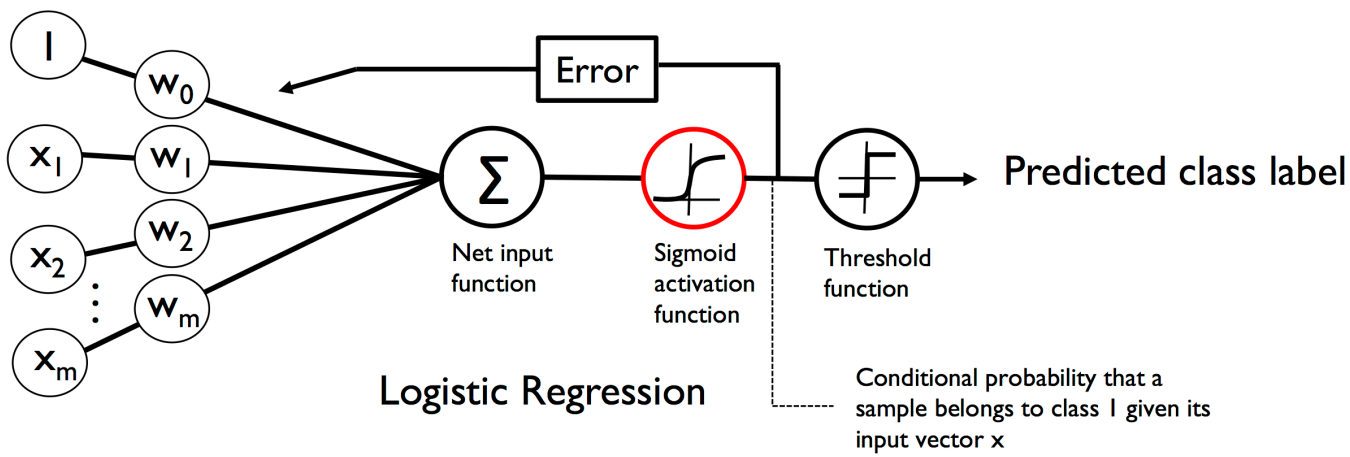
$$J(\beta) = -\sum_{i=1}^{M} \left[y^{(i)} \log\left(\sigma(z^{(i)})\right) + (1 - y^{(i)}) \log(1 - \sigma\left(z^{(i)}\right)) \right]$$

- Gradient

$$\nabla_\beta J(\beta) = -\sum_{i=1}^{M} \left(y^{(i)} - \sigma\left(z^{(i)}\right) \right) x^{(i)}$$

Logistic Regression Implementation





Logistic Regression Model

- For K classes, $p_k(\boldsymbol{x}), k=1,\cdots,K$ and $\sum_{k=1}^{K} p_k(\boldsymbol{x}) = 1$
  - Let $z_k = \beta_{k0} + \beta_k^T \boldsymbol{x},\ k = 1, \cdots K-1$ and $z_K = 0$
  - Define Softmax function

$$s(z_k) = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}},\ k=1,\cdots,K$$

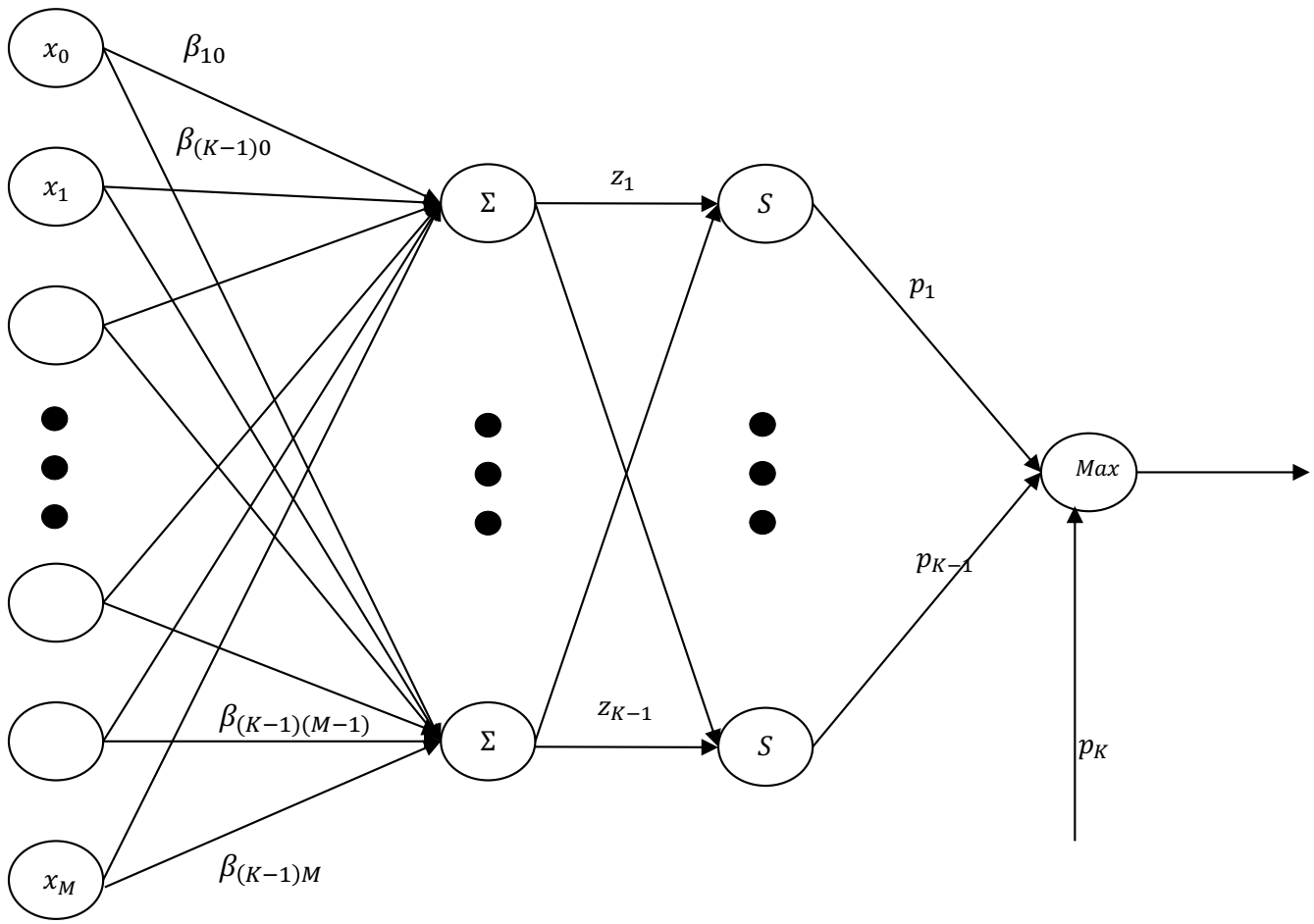
- Assume a model family

$$p_k(\boldsymbol{x}) = s(z_k)$$
$$= \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}, k = 1, \cdots, K$$
$$\sum_{k=1}^{K} p_k(\boldsymbol{x}) = 1$$

- Property

$$\log \frac{p_k(\boldsymbol{x})}{p_K(\boldsymbol{x})} = \beta_{k0} + \beta_k^T \boldsymbol{x},\ k = 1, \cdots, K-1$$

Prediction with Multiple Classes



Training for Logistic Regression Model

- $K > 2$
- Log-likelihood for M observations

$$J(\beta) = -\sum_{i=1}^{M} \log p\left(y^{(i)} \middle| \boldsymbol{x}^{(i)}, \beta\right)$$

$$p_k\left(\boldsymbol{x}^{(i)}\right) = s(z_k^{(i)})$$

$$J(\beta) = -\sum_{i=1}^{M} \sum_{k=1}^{K} I_k(y^{(i)}) \log s(z_k^{(i)})$$

where $I_k(.)$ is an indicator function

- Gradient

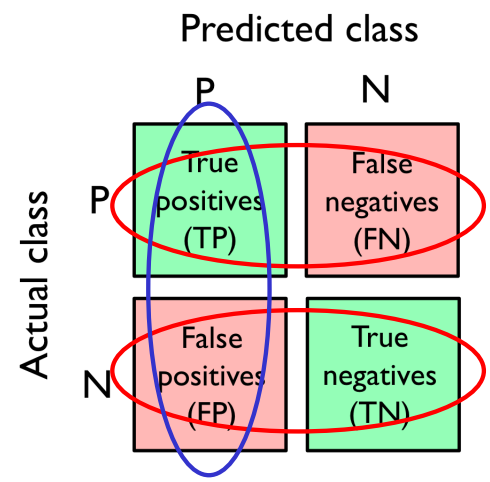
$$\nabla_{\beta_k} J(\beta_k) = -\sum_{i=1}^{M} \sum_{k=1}^{K} I_k(y^{(i)})(1 - s(z_k^{(i)}))\boldsymbol{x}^{(i)}$$

Performance Metrics

- Confusion matrix

$$ERR = \frac{FP+FN}{FP+FN+TP+TN}$$

$$ACC = \frac{TP+TN}{FP+FN+TP+TN} = 1 - ERR$$



- Imbalanced class
  - Ex. $P = 2, N = 98, TP = 1, FP = 2$

$FN = 1, TN = 96, ERR = 3\%, ACC = 97\%$

$P(X = 1) \approx 2\%, P(X = 0) \approx 98\%$

a priori

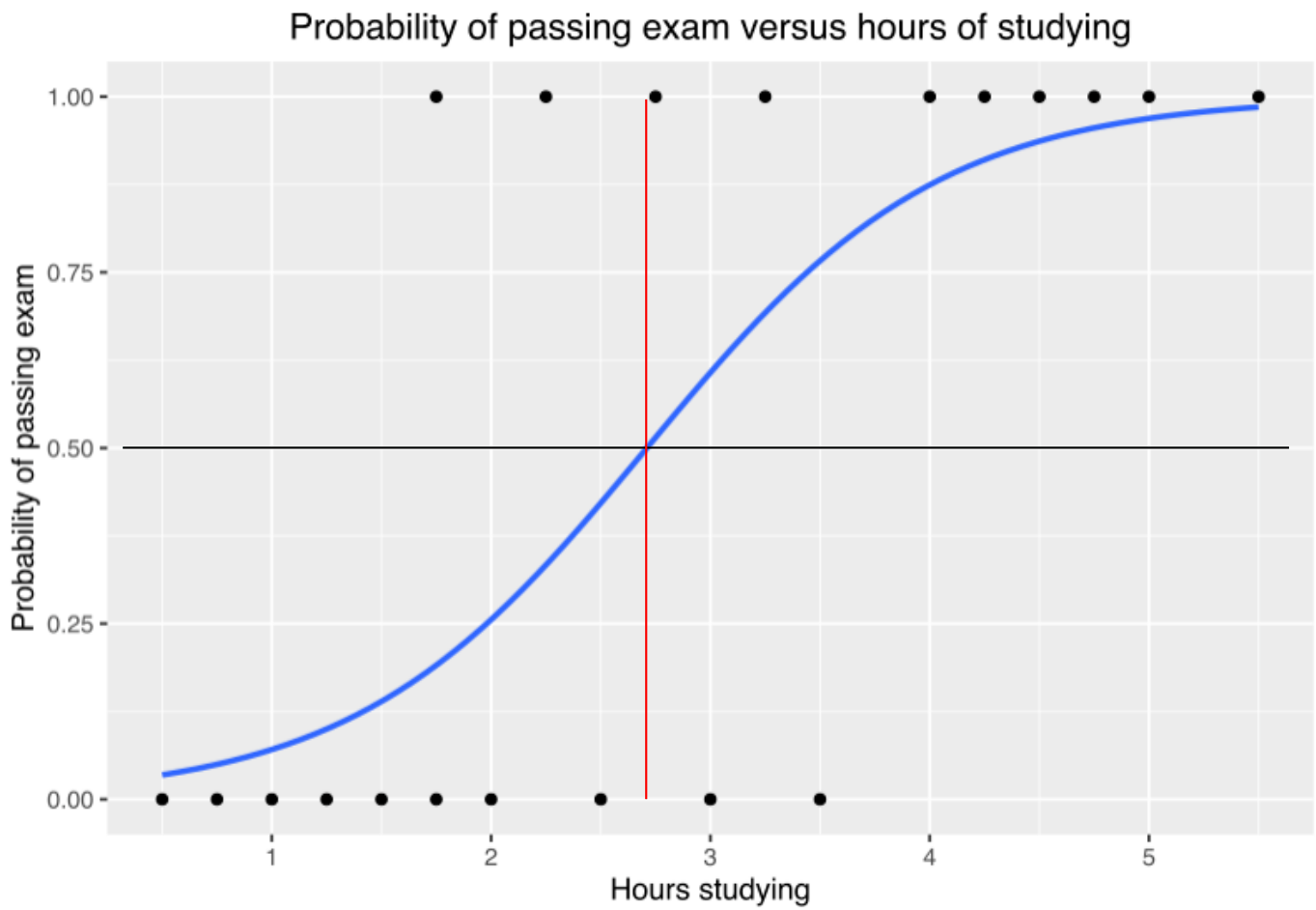
$P(Y = 1|X = 1) = \text{true positive rate}(sensitivity, recall) = \frac{TP}{TP+FN} = 50\%$

$P(Y = 0|X = 0) = \text{true negative rate}(specificity) = \frac{TN}{FP+TN} \approx 98\%$

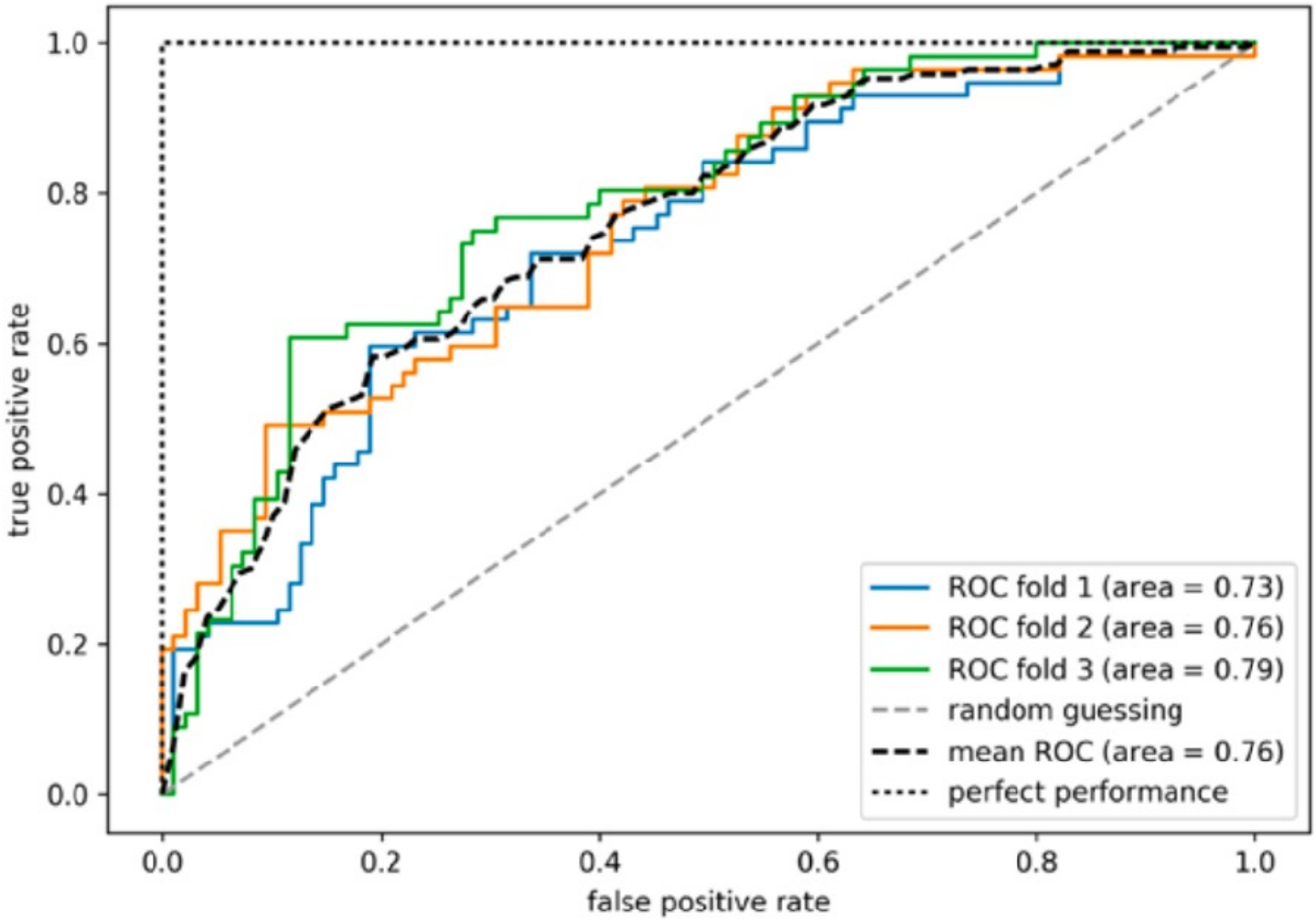
$P(X = 1|Y = 1) = \text{postive predictive value}(precision) = \frac{TP}{TP+FP} \approx 33.3\%$

a posteriori

Impacts of Threshold



Receiver Operating Characteristic Curve (ROC Curve)



Entropy

- A useful criterion for measuring information
- What is information?
  - Differentiate things
- How to measure information?
  - Rarer events carry more info

$$I(x) = -\log p(x)$$

- Definition

$$H(p) = E[I(X)] = -\sum_{x} p(x) \log p(x)$$

  - Average information an RV carries
  - Indicates uncertainness
  - Convention: $0 \log 0 = 0$

Cross-Entropy

- Average number of bits needed to encode data coming from a source with distribution P when we use model Q

$$H(P,Q) = -\sum_x p(x)\log q(x)$$

- $p(x)$ unknown, use examples to estimate

$$p(x)\log q(x) \approx \frac{1}{M}\left(\sum_{i=1}^{M} I_x(x_i)\right)\log q(x) = \frac{1}{M}\left(\sum_{i=1}^{M} I_x(x_i)\log q(x)\right)$$

- An example

$$x: \{0,1,1,2\} \qquad q(x) = \begin{cases} 0.2, x = 0 \\ 0.6, x = 1 \\ 0.2, x = 2 \end{cases}$$

$$H(P,Q) \approx \frac{1}{4}(-1\times\log 0.2 - 1\times\log 0.6 - 1\times\log 0.6 - 1\times\log 0.2) \approx 1.53$$

$$H(P) \approx -0.25\times\log 0.25 - 0.5\times\log 0.5 - 0.25\times\log 0.25 = 1.5$$

Measuring Loss

- Using negative log-likelihood

$$J(\beta) = -\sum_{i=1}^{M}\sum_{k=1}^{K} I_k(y^{(i)}) \log s(z_k^{(i)})$$

- Same as cross-entropy
  - Exampe $P(x=1) = 0.02, Q(x=1) = 0.03$

$$H(P,Q) = -0.02\times \log 0.03 - 0.98\times \log 0.97 \approx 0.144$$

$$H(P) = -0.02\times \log 0.02 - 0.98\times \log 0.98 \approx 0.141$$

Summary

- Key concepts

  Errors: intrinsic randomness, bias, variance

  Entropy, cross-entropy
- Models
  - Logistic Regression
- Algorithm
  - MLE

References

- *Deep Learning*, I. Goodfellow, Y. Bengio, and A. Courville, MIT Press, 2016
- *Python Machine Learning,* S. Raschka and V. Mirjalili, Packt Publishing, 3rd Edition, 2019